

Part 1: Topic Detection and Summarization of News Articles

This notebook uses **LangChain** with a **local Ollama model** (`llama3.2` , a small language model) to analyze BBC News articles and produce, for each article:

1. **Topic Classification** — Business / Entertainment / Politics / Sport / Tech
2. **Summarization** — a 2-3 sentence summary
3. **Key Entity Extraction** — people, organizations, and locations mentioned

We use Ollama (rather than a hosted API like Groq) to run everything locally without hitting rate limits, per the assignment's suggestion, which also lets us process the full dataset for the bonus section.

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```
In [1]: import json
import threading
from concurrent.futures import ThreadPoolExecutor
from pathlib import Path

import pandas as pd
from typing import List, Literal
from pydantic import BaseModel, Field
from tqdm.auto import tqdm

from langchain_ollama import ChatOllama
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser, JsonOutputParser

MODEL_NAME = "llama3.2" # local SLM served via Ollama
OLLAMA_URL = "http://localhost:11434" # explicit address: an OLLAMA_HOST of 0.0.0.0 is not a valid client address
# num_predict caps each reply so a looping generation cannot stall a long run
llm = ChatOllama(model=MODEL_NAME, temperature=0, base_url=OLLAMA_URL, num_predict=512)

print(f"Using Ollama model: {MODEL_NAME}")
```

Using Ollama model: llama3.2

Step 1: Load the Dataset

The BBC News Full-Text dataset (2,225 articles across 5 categories: business, entertainment, politics, sport, tech).

```
In [2]: raw_df = pd.read_csv("data/bbc-news-data.csv", sep="\t")
print(raw_df.shape)
print(raw_df["category"].value_counts())
raw_df.head(3)
```

```
(2225, 4)
category
sport      511
business   510
politics   417
tech       401
entertainment 386
Name: count, dtype: int64
```

```
Out[2]:
```

	category	filename	title	content
0	business	001.txt	Ad sales boost Time Warner profit	Quarterly profits at US media giant TimeWarne...
1	business	002.txt	Dollar gains on Greenspan speech	The dollar has hit its highest level against ...
2	business	003.txt	Yukos unit buyer faces loan claim	The owners of embattled Russian oil giant Yuk...

```
In [3]: # Build a clean working dataframe and limit to the first 30 articles
df = raw_df.rename(columns={"content": "Article_Text", "title": "Title", "category": "Category"}).copy()
df.insert(0, "Article_ID", df.index)
df = df[["Article_ID", "Title", "Category", "Article_Text"]]

df = df.head(30).reset_index(drop=True)
print(df.shape)
df.head()
```

```
(30, 4)
```

Out[3]:	Article_ID	Title	Category	Article_Text
0	0	Ad sales boost Time Warner profit	business	Quarterly profits at US media giant TimeWarne...
1	1	Dollar gains on Greenspan speech	business	The dollar has hit its highest level against ...
2	2	Yukos unit buyer faces loan claim	business	The owners of embattled Russian oil giant Yuk...
3	3	High fuel prices hit BA's profits	business	British Airways has blamed high fuel prices f...
4	4	Pernod takeover talk lifts Domecq	business	Shares in UK drinks and food firm Allied Dome...

Step 2: Topic Classification

We build a LangChain prompt (with a couple of few-shot examples) and force the model to return a single structured label using `with_structured_output`, so we never have to parse free text.

```
In [4]: class TopicClassification(BaseModel):
        topic: Literal["Business", "Entertainment", "Politics", "Sport", "Tech"] = Field(
            description="The single best-fitting news category for the article"
        )

classification_prompt = ChatPromptTemplate.from_messages([
    ("system",
     "You are a news editor. Analyze the given news article and identify its topic as one of the "
     "following categories: Business, Entertainment, Politics, Sport, or Tech. "
     "Respond with only the single best-fitting category.\n"
     "Business: companies, earnings, markets, the economy, trade, jobs, prices and interest rates.\n"
     "Entertainment: films, cinema, box office, actors, music, singers, albums, TV shows, theatre, books and awards.\n"
     "Politics: government, elections, parliament, political parties, ministers and laws.\n"
     "Sport: athletes, teams, clubs, matches, tournaments and championships.\n"
     "Tech: technology, computers, the internet, software, video games, mobile phones and telecoms.\n\n"
     "Examples:\n"
     "Article: 'Shares in the company jumped 5% after strong quarterly earnings were reported to investors.'\n"
     "Category: Business\n\n"
     "Article: 'The actress won an award for her role in the film, which also topped the box office.'\n"
     "Category: Entertainment\n\n"
     "Article: 'MPs debated the new immigration bill as the prime minister prepared for the election.'\n"
     "Category: Politics\n\n")
])
```

```

    "Article: 'The striker scored a hat-trick as his team won the league final 4-2 in extra time.'\n"
    "Category: Sport\n\n"
    "Article: 'The firm launched a new smartphone and a faster broadband service for gamers.'\n"
    "Category: Tech"),
    ("human", "Article:\n{article}\n\nCategory:")
])

classification_chain = classification_prompt | llm.with_structured_output(TopicClassification)

# --- Test on a sample datapoint ---
sample = df.iloc[0]
sample_result = classification_chain.invoke({"article": sample["Article_Text"][:3000]})
print("Title:", sample["Title"])
print("Actual category (dataset label):", sample["Category"])
print("Predicted topic:", sample_result.topic)

```

Title: Ad sales boost Time Warner profit
 Actual category (dataset label): business
 Predicted topic: Business

Step 3: Summarization

A simple prompt asking for a concise 2-3 sentence summary capturing who/what/when/where/why, without personal commentary.

```

In [5]: summarization_prompt = ChatPromptTemplate.from_messages([
    ("system",
     "You are a professional news summarizer. Summarize the main points of the given news article "
     "in 2-3 sentences, capturing who/what/when/where/why as applicable. "
     "Do not add personal commentary or opinions. Return only the summary text."),
    ("human", "Article:\n{article}")
])

summarization_chain = summarization_prompt | llm | StrOutputParser()

# --- Test on a sample datapoint ---
sample_summary = summarization_chain.invoke({"article": sample["Article_Text"][:4000]})
print("Title:", sample["Title"])
print("Summary:", sample_summary)

```

Title: Ad sales boost Time Warner profit

Summary: Time Warner's quarterly profits jumped 76% to \$1.13 billion, driven by sales of high-speed internet connections and higher advert sales, with the company now owning 8% of search engine Google. The firm's fourth quarter sales rose 2% to \$11.1 billion, with its internet business AOL experiencing a mixed fortune, losing 464,000 subscribers but seeing underlying profit rise 8%. Time Warner is re-stating its 2000 and 2003 results following a probe by the US Securities Exchange Commission.

Step 4: Key Entity Extraction

We extract entities using **JSON-mode prompting** (Ollama's `format="json"`) with a Pydantic schema for validation via `JsonOutputParser`. In testing, LangChain's tool-calling `with_structured_output` frequently returned empty entity lists on longer articles with this small model — JSON-mode prompting proved much more reliable, so we use it here instead.

```
In [6]: class KeyEntities(BaseModel):
    people: List[str] = Field(default_factory=list, description="Notable people mentioned in the article")
    organizations: List[str] = Field(default_factory=list, description="Organizations/companies mentioned")
    locations: List[str] = Field(default_factory=list, description="Places/locations mentioned")

entity_parser = JsonOutputParser(pydantic_object=KeyEntities)

entity_prompt = ChatPromptTemplate.from_messages([
    ("system",
     "From the article below, list the names of any important people, organizations, or places mentioned. "
     "If a category has no entities, return an empty list for it.\n{format_instructions}"),
    ("human", "Article:\n{article}")
]).partial(format_instructions=entity_parser.get_format_instructions())

# JSON-mode (format="json") makes the Ollama model emit valid JSON directly, which this small
# model handles more reliably than LangChain's tool-calling with_structured_output.
entity_llm = ChatOllama(model=MODEL_NAME, temperature=0, format="json", base_url=OLLAMA_URL, num_predict=512)
entity_chain = entity_prompt | entity_llm | entity_parser

# --- Test on a sample datapoint ---
sample_entities = entity_chain.invoke({"article": sample["Article_Text"][:4000]})
print("Title:", sample["Title"])
print("Entities:", sample_entities)
```

Title: Ad sales boost Time Warner profit

Entities: {'people': ['Richard Parsons'], 'organizations': ['TimeWarner', 'Google', 'Warner Bros', 'AOL', 'SEC', 'Bertelsmann'], 'locations': []}

Step 5: Apply to All Articles and Update the DataFrame

```
In [7]: def analyze_article(article_text: str) -> dict:
    text = article_text[:4000] # keep prompts within a reasonable context size for the SLM
    try:
        topic = classification_chain.invoke({"article": text}).topic
    except Exception:
        topic = "Unknown"
    try:
        summary = summarization_chain.invoke({"article": text}).strip()
    except Exception:
        summary = ""
    try:
        entities = entity_chain.invoke({"article": text})
    except Exception:
        entities = {"people": [], "organizations": [], "locations": []}
    return {"Detected_Topic": topic, "Summary": summary, "Key_Entities": entities}

def run_pipeline(data: pd.DataFrame, cache_path: str | None = None, workers: int = 1) -> pd.DataFrame:
    """Analyze every article. With `cache_path`, each finished row is saved to a JSON-lines file so a
    long run (the bonus) can resume where it stopped. Rows where a chain failed are retried once.
    `workers` > 1 sends several articles to Ollama at once (it serves requests in parallel)."""
    done = {}
    if cache_path and Path(cache_path).exists():
        for line in Path(cache_path).read_text(encoding="utf-8").splitlines():
            record = json.loads(line)
            done[record["Article_ID"]] = record["result"]
    todo = [(int(row["Article_ID"]), row["Article_Text"]) for _, row in data.iterrows()
            if int(row["Article_ID"]) not in done]
    cache = open(cache_path, "a", encoding="utf-8") if cache_path else None
    lock = threading.Lock()

    def work(item):
        key, text = item
        result = analyze_article(text)
        if result["Detected_Topic"] == "Unknown" or not result["Summary"]:
            result = analyze_article(text) # retry once
        with lock:
            done[key] = result
```

```

        if cache:
            cache.write(json.dumps({"Article_ID": key, "result": result}) + "\n")
            cache.flush()

    with ThreadPoolExecutor(max_workers=workers) as pool:
        list(tqdm(pool.map(work, todo), total=len(todo), desc="Analyzing articles"))
    if cache:
        cache.close()
    results = [done[int(k)] for k in data["Article_ID"]]
    results_df = pd.DataFrame(results)
    return pd.concat([data.reset_index(drop=True), results_df], axis=1)

final_df = run_pipeline(df)
final_df.head()

```

Out[7]:

	Article_ID	Title	Category	Article_Text	Detected_Topic	Summary	Key_Entities
0	0	Ad sales boost Time Warner profit	business	Quarterly profits at US media giant TimeWarne...	Business	Time Warner's quarterly profits jumped 76% to ...	{'people': ['Richard Parsons'], 'organizations': ...}
1	1	Dollar gains on Greenspan speech	business	The dollar has hit its highest level against ...	Business	The US dollar has reached its highest level ag...	{'people': ['Alan Greenspan', 'Robert Sinche']...}
2	2	Yukos unit buyer faces loan claim	business	The owners of embattled Russian oil giant Yuk...	Business	The owners of Yukos, Menatep Group, plan to as...	{'people': ['Mikhail Khodorkovsky', 'Jamie Fir...']}
3	3	High fuel prices hit BA's profits	business	British Airways has blamed high fuel prices f...	Business	British Airways reported a 40% drop in profits...	{'people': ['Rod Eddington', 'Mike Powell', 'M...']}
4	4	Pernod takeover talk lifts Domecq	business	Shares in UK drinks and food firm Allied Dome...	Business	Shares in UK drinks and food firm Allied Domec...	{'people': ['Pernod Ricard'], 'organizations': ...}

```

In [8]: # Quick sanity check: how often does the detected topic match the dataset's original label?
accuracy = (final_df["Detected_Topic"].str.lower() == final_df["Category"].str.lower()).mean()
print(f"Topic classification accuracy vs. ground-truth label (first 30 articles): {accuracy:.2%}")

```

Topic classification accuracy vs. ground-truth label (first 30 articles): 83.33%

```

In [9]: final_df.to_csv("outputs/part1_bbc_first30_results.csv", index=False)
final_df.to_json("outputs/part1_bbc_first30_results.json", orient="records", indent=2)
print("Saved to outputs/part1_bbc_first30_results.csv and .json")
final_df

```

Saved to outputs/part1_bbc_first30_results.csv and .json

Out[9]:

	Article_ID	Title	Category	Article_Text	Detected_Topic	Summary	Key_Entities
0	0	Ad sales boost Time Warner profit	business	Quarterly profits at US media giant TimeWarne...	Business	Time Warner's quarterly profits jumped 76% to ...	{'people': ['Richard Parsons'], 'organizations': ...}
1	1	Dollar gains on Greenspan speech	business	The dollar has hit its highest level against ...	Business	The US dollar has reached its highest level ag...	{'people': ['Alan Greenspan', 'Robert Sinche']...}
2	2	Yukos unit buyer faces loan claim	business	The owners of embattled Russian oil giant Yuk...	Business	The owners of Yukos, Menatep Group, plan to as...	{'people': ['Mikhail Khodorkovsky', 'Jamie Fir...']}
3	3	High fuel prices hit BA's profits	business	British Airways has blamed high fuel prices f...	Business	British Airways reported a 40% drop in profits...	{'people': ['Rod Eddington', 'Mike Powell', 'M...']}
4	4	Pernod takeover talk lifts Domecq	business	Shares in UK drinks and food firm Allied Dome...	Business	Shares in UK drinks and food firm Allied Domec...	{'people': ['Pernod Ricard'], 'organizations':...}
5	5	Japan narrowly escapes recession	business	Japan's economy teetered on the brink of a te...	Business	Japan's economy experienced a technical recess...	{'people': ['Heizo Takenaka', 'Paul Sheard'], ...}
6	6	Jobs growth still slow in the US	business	The US created fewer jobs than expected in Ja...	Business	The US economy added 146,000 jobs in January, ...	{'people': ['Rick Egelton', 'Herbert Hoover', ...]}
7	7	India calls for fair trade rules	business	India, which attends the G7 meeting of seven ...	Politics	India's finance minister, Palaniappan Chidamba...	{'people': ['Palaniappan Chidambaram', 'Gordon...']}
8	8	Ethiopia's crop production up 24%	business	Ethiopia produced 14.27 million tonnes of cro...	Business	Ethiopia's crop production increased by 21% in...	{'people': ['Henri Josserand'], 'organizations': ...}
9	9	Court rejects \$280bn tobacco case	business	A US government claim accusing the country's ...	Politics	A US appeals court has rejected a \$280bn claim...	{'people': ['Clinton'], 'organizations': ['Alt...']}
10	10	Ask Jeeves tips online ad revival	business	Ask Jeeves has become the third leading onlin...	Business	Ask Jeeves' revenue nearly tripled in the four...	{'people': ['Ask Jeeves'], 'organizations': ['...']}
11	11	Indonesians face fuel price rise	business	Indonesia's government has confirmed it is co...	Politics	Indonesia's government is considering raising ...	{'people': ['Susilo Bambang Yudhoyono', 'Megaw...']}
12	12	Peugeot deal boosts Mitsubishi	business	Struggling Japanese car maker Mitsubishi Moto...	Business	Mitsubishi Motors has agreed to supply Peugeot...	{'people': ['Takashi Nishioka'], 'organization': ...}
13	13	Telegraph newspapers axe 90 jobs	business	The Daily and Sunday Telegraph newspapers are...	Business	The Daily and Sunday Telegraph newspapers are ...	{'people': ['Jeremy Dear', 'Barry Fitzpatrick']...}
14	14	Air passengers win new EU rights	business	Air passengers who are unable to board their ...	Business	The European Union has introduced new rules re...	{'people': ['Michael Cawley', 'Jacques Barrot']...}

	Article_ID	Title	Category	Article_Text	Detected_Topic	Summary	Key_Entities
15	15	China keeps tight rein on credit	business	China's efforts to stop the economy from over...	Business	China's government plans to continue credit co...	{'people': [''], 'organizations': [''], 'locat...
16	16	Parmalat boasts doubled profits	business	Parmalat, the Italian food group at the centr...	Business	Parmalat, the Italian food group, has reported...	{'people': ['Enrico Bondi'], 'organizations': ...
17	17	India's rupee hits five-year high	business	India's rupee has hit a five-year high after ...	Business	India's rupee has reached a five-year high of ...	{'people': ['Bhanu Baweja'], 'organizations': ...
18	18	India widens access to telecoms	business	India has raised the limit for foreign direct...	Business	India has raised the foreign direct investment...	{'people': ['Dayanidhi Maran', 'Manmohan Singh...
19	19	Call centre users 'lose patience'	business	Customers trying to get through to call centr...	Business	A survey suggests that customers are becoming ...	{'people': ['Cara Diemont'], 'organizations': ...
20	20	Rank 'set to sell off film unit'	business	Leisure group Rank could unveil plans to deme...	Business	Rank, a leisure group formerly famous for the ...	{'people': ['Mike Smith'], 'organizations': ['...
21	21	Sluggish economy hits German jobs	business	The number of people out of work in Europe's ...	Business	German unemployment rose 7,000 in November to ...	{'people': ['Frank-Juergen Weise', 'Gerhard Sc...
22	22	Mixed signals from French economy	business	The French economy picked up speed at the end...	Business	The French economy experienced a 0.7-0.8% grow...	{'people': ['David Naude'], 'organizations': [...
23	23	US trade gap hits record in 2004	business	The gap between US exports and imports hit an...	Business	The US trade deficit reached an all-time high ...	{'people': ['President Bush', 'Marie-Pierre Ri...
24	24	Yukos loses US bankruptcy battle	business	A judge has dismissed an attempt by Russian o...	Politics	A US judge has dismissed Yukos' attempt to gai...	{'people': ['Mikhail Khodorkovsky', 'Letitia C...
25	25	Safety alert as GM recalls cars	business	The world's biggest carmaker General Motors (...)	Business	General Motors is recalling nearly 200,000 veh...	{'people': ['General Motors'], 'organizations'...
26	26	Steel firm 'to cut' 45,000 jobs	business	Mittal Steel, one of the world's largest stee...	Business	Mittal Steel, the Netherlands-based steel prod...	{'people': ['Lakshmi Mittal'], 'organizations'...
27	27	Strong demand triggers oil rally	business	Crude oil prices surged back above the \$47 a ...	Business	Crude oil prices surged above \$47 a barrel on ...	{'people': ['Orin Middleton'], 'organizations'...
28	28	UK firm faces Venezuelan land row	business	Venezuelan authorities have said they will se...	Politics	Venezuelan authorities plan to seize 12,900 ac...	{'people': ['Alexis Ortiz'], 'organizations': ...
29	29	Soaring oil 'hits world economy'	business	The soaring cost of oil has hit global econom...	Business	The OECD has cut its growth predictions for ma...	{'people': ['Jean-Philippe Cotis', 'Vincent Ko...

Bonus: Full Dataset (all 2,225 articles)

`RUN_BONUS = True` below processes the *entire* dataset instead of just the first 30. Since this runs a local SLM via Ollama with no external API rate limits, it is feasible, but it will take a while (roughly 1-2 hours depending on hardware, since each article needs 3 LLM calls: classification, summarization, entity extraction).

Finished rows are saved to `outputs/part1_bbc_ALL_cache.jsonl` as the run goes, so an interrupted run continues where it stopped. Set `RUN_BONUS = False` to skip this cell.

```
In [10]: RUN_BONUS = True

if RUN_BONUS:
    bonus_df = raw_df.rename(columns={"content": "Article_Text", "title": "Title", "category": "Category"}).copy()
    bonus_df.insert(0, "Article_ID", bonus_df.index)
    bonus_df = bonus_df[["Article_ID", "Title", "Category", "Article_Text"]]

    bonus_results_df = run_pipeline(bonus_df, cache_path="outputs/part1_bbc_ALL_cache.jsonl", workers=4)
    bonus_results_df.to_csv("outputs/part1_bbc_ALL_results.csv", index=False)
    bonus_results_df.to_json("outputs/part1_bbc_ALL_results.json", orient="records", indent=2)

    bonus_accuracy = (bonus_results_df["Detected_Topic"].str.lower() == bonus_results_df["Category"].str.lower()).mean()
    print(f"Full-dataset classification accuracy: {bonus_accuracy:.2%}")
    failed = ((bonus_results_df["Detected_Topic"] == "Unknown") | (bonus_results_df["Summary"] == "")).sum()
    print(f"Rows analysed: {len(bonus_results_df)} | rows with a failed chain: {failed}")
    print("Saved to outputs/part1_bbc_ALL_results.csv and .json")
else:
    print("RUN_BONUS is False - skipping full-dataset run. Set RUN_BONUS = True to process all 2,225 articles.")
```

Full-dataset classification accuracy: 82.56%
Rows analysed: 2225 | rows with a failed chain: 0
Saved to outputs/part1_bbc_ALL_results.csv and .json